

A Quarto manuscript template for RevTeX

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This is a placeholder abstract. The document demonstrates the features of a Quarto manuscript project that renders a RevTeX PDF: author and affiliation metadata, numbered sections and equations, cross-references, citations, and figures generated from Python code.

I. WORKFLOW

This section documents how the manuscript is built and should be removed before submission. The source is the Quarto file `index.qmd`, and `_quarto.yml` configures a Quarto *manuscript* project that renders HTML, Word, JATS and a RevTeX PDF from that single source.

A. Building

Render all formats, or only the PDF, with

```
quarto render
quarto render index.qmd --to pdf
```

Output is written to `_manuscript/`. Because `freeze: true` is set, a project-wide render reuses stored code output; rendering `index.qmd` explicitly re-executes the Python chunks. Use `quarto preview` for live reloading while writing.

B. RevTeX setup

The PDF uses `documentclass: revtex4-2` with the options `aps`, `pra`, `reprint`, `superscriptaddress`; change `pra` to target another journal, or `reprint` to preprint for a single-column draft. The folder `_revtex/` holds the glue between Quarto and RevTeX. The file `before-body.tex` emits `\title`, `\author`, `\affiliation`, `\email` and the abstract after `\begin{document}`, as RevTeX requires. The file `preamble.tex` resolves a `longtable*` clash with the caption package. The generated LaTeX is kept in `_manuscript/_tex/` for journal submission.

C. Fonts

Text and mathematics are set in STIX Two in both the PDF and the HTML output. The font files are in `_fonts/`, and they are switched on by `stix: true` in `_quarto.yml`;

set it to `false` to use the default fonts. The filter `_fonts/stix.lua` loads the fonts in the PDF preamble and, for HTML, tells MathJax 4 to use its STIX Two math font.

D. Plots

The first code chunk in this file loads the local package `revtexplot`, which styles matplotlib figures to match the PDF. It is switched on by `plot-style: true` in `_quarto.yml`; set it to `false` for matplotlib's defaults. Figures are one column wide by default and use 8 pt text, the size of `\footnotesize`, in STIX Two (or Computer Modern with `stix: false`). They are saved at exactly their set size, so the text in them matches the captions. The package provides these sizes:

- `rp.COLUMN`: one column, 3.40×2.10 in (the default)
- `rp.WIDE`: both columns, 7.06×2.35 in
- `rp.figsize(width, aspect)`: width is "column", "wide" or inches, and aspect is height / width

For example, `plt.subplots(figsize=rp.figsize("column", 1))` gives a square one-column figure.

E. Writing

Authors, affiliations, e-mail addresses and the abstract are set in the YAML header. Sections, equations and figures get labels with the prefixes `sec-`, `eq-` and `fig-`, and are referenced as `@sec-methods`, `@eq-gaussian` or `@fig-example`. A displayed equation is labelled by appending `{#eq-name}` after its closing `$$`. Citations use keys from `references.bib`, e.g. `[@knuth1984]`, and are formatted by BibTeX with the `apsrev4-2` style. Figures are produced by Python chunks whose `label` and `fig-cap` options set the reference and caption; see Section IV for an example.

F. Wide figures and equations

In the two-column reprint layout, a figure spans both columns when it uses RevTeX's `figure*` environment. For a Python chunk, add the options

```
#| fig-env: "figure*"
#| fig-pos: "t"
```

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and create the figure with `plt.subplots(figsize=rp.WIDE)` (see Section ID). For an image file, write

```
![[Caption.](figures/wide.pdf){#fig-wide
fig-env="figure*" fig-pos="t"}]
```

Wide figures float to the top of a following page. Always set `fig-pos: "t"`: Quarto places chunk figures with [H] by default, which makes a `figure*` vanish from the PDF. Do not use `column: page`; it loads the `sidenotes` package, which stops every `figure*` from spanning both columns.

A long equation is set across both columns by wrapping it in RevTeX's `widetext` environment with raw LaTeX blocks:

```
```{=latex}
\begin{widetext}
```
$$
E = \sum_n a_n x^n + \dots
$$ {#eq-wide}
```{=latex}
\end{widetext}
```
```

Labels and cross-references work as usual. The HTML output ignores `fig-env`, `fig-pos` and the raw LaTeX, so both render normally there.

II. INTRODUCTION

This is placeholder text for the introduction. Literate programming [1] advocates weaving code and prose into a single document, an idea that Quarto applies to scientific manuscripts. The typesetting itself is done by TeX [2] and LaTeX [3]. Multiple references can be cited together [1, 3].

III. METHODS

A. Inline and displayed mathematics

Inline mathematics is written between single dollar signs, as in $e^{i\pi} + 1 = 0$. Displayed equations are written between double dollar signs and numbered when given a label, as for the Gaussian integral

$$\int_{-\infty}^{\infty} e^{-ax^2} dx = \sqrt{\frac{\pi}{a}}, \quad a > 0. \quad (1)$$

B. Multi-line equations

Long expressions can be split with an `aligned` environment inside a single labelled equation,

$$\begin{aligned} f(x) &= \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n \\ &= f(x_0) + f'(x_0)(x - x_0) + \mathcal{O}((x - x_0)^2). \end{aligned} \quad (2)$$

C. Cross-references

Equations are referenced as Equation 1 and Equation 2, sections as Section IV, and figures as Figure 1. The numbers are assigned automatically in every output format.

IV. RESULTS

Figure 1 is generated by the Python code chunk embedded in the source file. It shows the damped oscillation

$$f(t) = e^{-\gamma t} \cos(\omega t). \quad (3)$$

for three values of the damping rate γ .

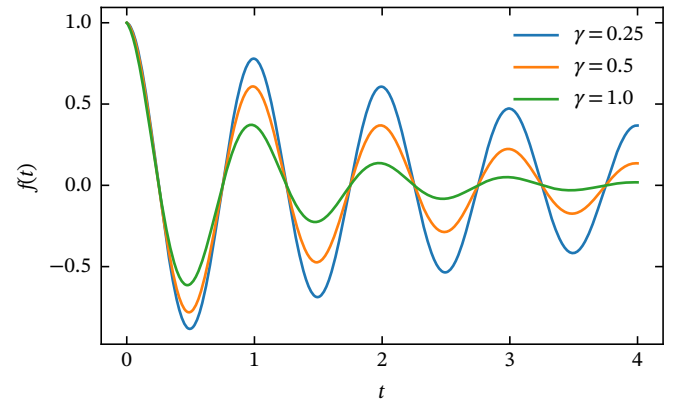


Figure 1: Damped oscillations for three values of the damping rate γ , with $\omega = 2\pi$.

Figure 2 spans both columns and shows each damping rate in its own panel.

V. CONCLUSION

This is placeholder text for the conclusion. Replace the contents of this file with your own manuscript, keeping Section I as a reference until submission.

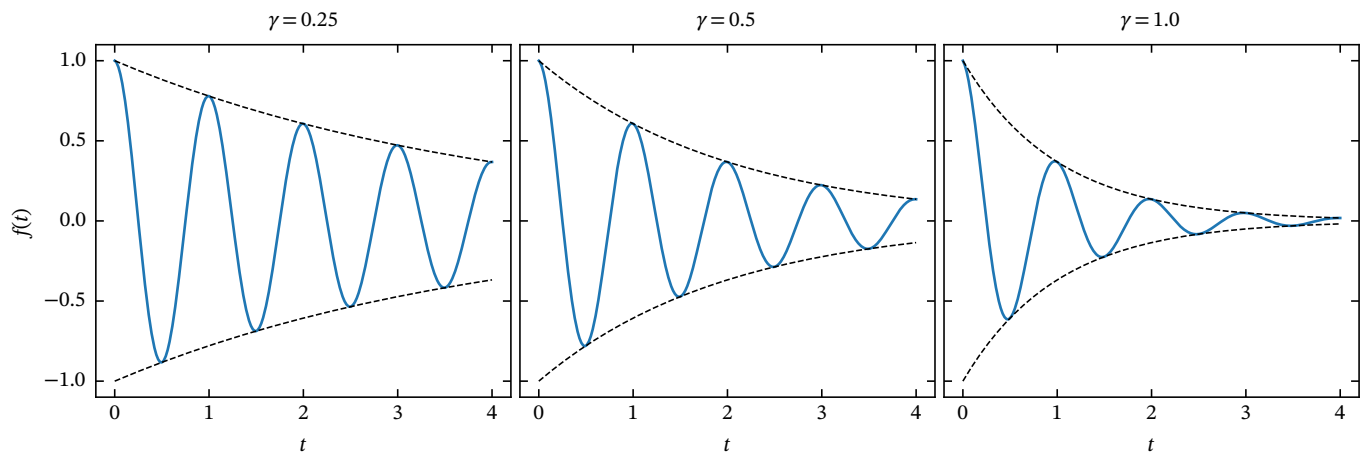


Figure 2: The oscillations of Figure 1 with their envelopes $\pm e^{-\gamma t}$ (dashed).

[1] D. E. Knuth, [Comput. J.](#) **27**, 97 (1984).

[2] D. E. Knuth, *The T_EXbook* (Addison-Wesley, Reading, MA, 1986).

[3] L. Lamport, *L^AT_EX: A Document Preparation System*, 2nd ed. (Addison-Wesley, Reading, MA, 1994).